# Package 'ssGSEA3'

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**Title** Single-Sample and Site-Centric Gene Set Enrichment Analysis (ssGSEA/PTM-SEA)

**Version** 1.0.0

**Description** Tools for gene-centric (ssGSEA) and site-centric (PTM-SEA) single-sample gene set enrichment analysis: GCT preprocessing into gene-centric or single-site-centric reports, a gene-level bootstrap enrichment/significance engine, and summary heatmap reporting with HALLMARK and PTMSigDB term groupings.

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`hallmark_categories` *HALLMARK gene set process categories*

---

## Description

Named character vector mapping each of the 50 MSigDB HALLMARK gene set names to a broad process category (signaling, immune, metabolic, etc.), used to color-code and group rows in summary heatmaps.

## Usage

```
hallmark_categories
```

## Format

A named character vector of length 50; names are HALLMARK gene set IDs, values are one of "signaling", "immune", "development", "proliferation", "cellular component", "metabolic", "DNA damage", or "pathway".

## Source

Broad Institute MSigDB HALLMARK collection; category assignments originate from the lab's ssGSEA reporting scripts (`pw_hm()`).

## hallmark_category_colors *Color palette for HALLMARK process categories*

### Description

Color palette for HALLMARK process categories

### Usage

```
hallmark_category_colors
```

### Format

A named character vector of hex color codes, one per category value in hallmark_categories.

## plot_rank_diagnostics *Rank ("mountain") plot diagnostic for ssGSEA/PTM-SEA results*

### Description

For each selected gene set, draws one page with one panel per sample: the sample's original expression values, ranked high to low, with the gene set's members marked along that ranking. This is the classic GSEA enrichment-plot diagnostic, adapted to ssGSEA/PTM-SEA – useful for visually confirming that a significant score is actually driven by members clustered toward one end of the ranking, not a fluke.

### Usage

```
plot_rank_diagnostics(
  result,
  terms = NULL,
  top_n = 10,
  direction = c("both", "pos", "neg"),
  fdr.max = 0.05,
  samples = NULL,
  file = "rank-plots.pdf",
  mfrow = c(3, 3)
)
```

### Arguments

| | |
|---|---|
| result | An ssgsea_result, as returned by ssGSEA3(). |
| terms | Character vector of gene set names to plot. If NULL (default), gene sets are chosen via select_top_terms() using top_n/ direction/fdr.max. |
| top_n, direction, fdr.max | Passed to select_top_terms() when terms is NULL. |
| samples | Character vector of sample names to plot (as panels on each gene set's page). Defaults to all samples in result. |

| | |
|---|---|
| file | Path to write a (possibly multi-page) PDF to. If NULL, plots to the current graphics device instead (one page per term). |
| mfrow | Panel layout (rows, cols) per page; samples beyond prod(mfrow) for a single gene set spill onto additional pages. |

## Details

Unlike the original implementation (which re-parsed the written score/ p-value/FDR GCT files back off disk for every plot), this draws directly from the result object already in memory.

## Value

(Invisibly) the character vector of gene sets plotted.

---

# plot_ssgsea_heatmap *ssGSEA/PTM-SEA summary heatmap*

---

## Description

Reworks the lab's existing pw_hm() heatmap (base pheatmap) onto ComplexHeatmap. Selects gene sets via select_top_terms() by default (FDR-significant, then top-N by direction), colors a symmetric blue-white- red scale centered at zero, and marks FDR-significant cells with *.

## Usage

```
plot_ssgsea_heatmap(
  result,
  terms = NULL,
  top_n = 10,
  direction = c("both", "pos", "neg"),
  fdr.max = 0.05,
  samples = NULL,
  split_by_db_category = TRUE,
  cluster_rows = TRUE,
  cell.size = 10,
  title = NULL,
  nes.range = NULL,
  title_fontsize = 14,
  show_row_names = TRUE,
  show_category = TRUE,
  show_heatmap_legend = TRUE,
  show_row_dend = TRUE
)
```

## Arguments

| | |
|---|---|
| result | An ssgsea_result, as returned by ssGSEA3(). |
| terms | Character vector of gene set names to plot. If NULL (default), gene sets are chosen via select_top_terms() using top_n/ direction/fdr.max. |

top_n, direction, fdr.max
: Passed to select_top_terms() when terms is NULL; fdr.max is also the threshold used for marking significant cells with *.

samples
: Character vector of sample names (columns) to include. Defaults to all samples in result.

split_by_db_category
: If TRUE (default), auto-detect and split by a shared CATEGORY-name prefix (e.g. PTMSigDB) when HALLMARK grouping doesn't apply.

cluster_rows
: Controls row order/dendrogram; forwarded as-is to ComplexHeatmap::Heatmap()'s own cluster_rows, which accepts three different kinds of value:

  - TRUE (default): cluster fresh from the matrix being plotted. Within each group/slice when HALLMARK or prefix-based grouping applies, across all rows otherwise. Dendrogram lines are drawn.
  - FALSE: no clustering at all – rows are shown in exactly the order given in terms (or the order select_top_terms() returned).
  - A precomputed hclust/dendrogram object, or a function that takes the matrix being plotted and returns one: your clustering is used exactly as given, *without* being re-sorted to better fit this particular matrix's values (Heatmap()'s own row_dend_reorder otherwise does this by default, which is why this case turns it off automatically here). Useful for drawing several plot_ssgsea_heatmap() results side by side (e.g. comparing two pipeline versions) with identical row order/dendrogram across panels, even though each panel's underlying matrix differs.

cell.size
: Fixed cell width/height, in points, matching pw_hm()'s own cw/ch default of 10 – keeps cells square regardless of the number of rows/columns, rather than stretching to fill the plotting device (which is what a bare ComplexHeatmap::Heatmap() call does by default when width/height aren't set).

title
: Optional heatmap title.

nes.range
: Optional c(min, max) to fix the color scale instead of auto-computing it from the terms/samples actually being plotted. Pin this to the same values across several heatmaps – e.g. one showing all significant terms and another showing a hand-picked illustrative subset – so a subset with modest scores reads as visually modest rather than stretching to fill the color scale's extremes on its own narrower range. When computing this externally from a larger pool (e.g. a whole gene set database), do NOT use that pool's literal min/max – a single extreme outlier gene set (common in real data; e.g. a tiny gene set with an inflated score) will dominate the range and wash out every other heatmap pinned to it, including ones with genuinely large, real effect sizes. Use a percentile clamp instead (e.g. the 1st/99th percentile of NES among gene sets significant somewhere in that pool) so a handful of outliers can't blow out the shared scale.

title_fontsize
: Font size for title. Default 14 suits a standalone heatmap with several sample columns; narrow panels (e.g. 2 columns) may need a smaller size to avoid the title overflowing its own column width, particularly when concatenating several calls' results side by side.

show_row_names, show_category, show_heatmap_legend, show_row_dend
: Individually toggle the row names, category color bar/title, color-scale legend, and row dendrogram – all default TRUE for standalone use, but useful to selectively disable when concatenating several calls' results into one multi-panel comparison figure (e.g. category bar only on the first panel, row names only on the last, one shared legend).

### Details

Row grouping is auto-detected from the selected terms' names:

- If any term is a bundled hallmark_categories name, rows are split and colored by HALLMARK process category (matching pw_hm()'s behavior exactly), with the HALLMARK_ prefix stripped from row labels.
- Otherwise, if split_by_db_category and the terms carry a shared CATEGORY-name prefix convention (as in PTMSigDB: KINASE-, PERT-, PATH-, DISEASE-) with more than one distinct prefix present, rows are split by that prefix.

In every case, rows are hierarchically clustered by default (cluster_rows = TRUE) using ComplexHeatmap's own built-in clustering rather than pw_hm()'s seriation::seriate() – avoids an extra dependency for what's typically a very similar row order. When a grouping applies, clustering happens *within* each group/slice (slice order itself is left as the groups' first-encountered order, not reclustered). See the cluster_rows argument below for the two other values it accepts besides TRUE/FALSE.

### Value

A ComplexHeatmap::Heatmap object – draw it with ComplexHeatmap::draw(), or save with e.g. pdf()/png() around that call.

---

## preprocessGCT *Preprocess a GCT file for ssGSEA or PTM-SEA*

---

### Description

Converts a raw GCT file into one of three report levels:

- "gc" (gene-centric): collapses multiple rows per gene into one, per mode. **This is the level** ssGSEA3() **expects** – exactly one row per gene, matching its assumption that each gene contributes a single ranked value to the enrichment statistic.
- "ssc" (single-site-centric): the PTM-SEA path. Explodes multiply-modified peptides into one row per site and builds a PTM-SEA-compatible site ID – also one row per (site-level) scoring unit, so also directly usable by ssGSEA3().
- "gcr" (gene-centric-redundant): relabels rows with gene symbols but does *not* collapse – every original row is kept (deduplicating names only, e.g. GeneA, GeneA_1). This is intended as an inspection/reporting format (e.g. seeing every individual peptide/probe measurement under its gene symbol before choosing how to collapse it), **not** as ssGSEA3() input: feeding it multiple rows for the same gene in would each count as an independent "hit" in the enrichment ranking, giving genes with more rows disproportionate influence on the score for reasons unrelated to real biology. Use "gc" for ssGSEA scoring instead.

### Usage

```
preprocessGCT(
  gct.str,
  level = c("ssc", "gc", "gcr"),
  id.type = c("sm", "ph"),
  id.type.out = c("uniprot", "refseq", "seqwin"),
  acc.type = c("uniprot", "refseq", "symbol"),
```

```
  organism = NULL,
  seqwin.col = "VMsiteFlanks",
  gene.col = "geneSymbol",
  fix.gene.names = FALSE,
  fix.gene.names.column = "id.description",
  fix.gene.names.regexpr = ".* GN\\=(.*)$",
  humanize.gene.names = FALSE,
  loc = TRUE,
  mode = c("mean", "median", "sd", "SGT", "abs.max"),
  SGT.col = "subgroupNum",
  mod.res = c("S|T|Y", "K"),
  mod.type = c("p", "ac", "ub"),
  appenddim = TRUE
)
```

## Arguments

| | |
|---|---|
| `gct.str` | Path to the input GCT (v1.2/v1.3) file. |
| `level` | One of `"ssc"`, `"gc"`, `"gcr"` (see above). Use `"gc"` or `"ssc"` for data that will be scored with `ssGSEA3()`; `"gcr"` is not recommended for that purpose (see above). |
| `id.type` | Input site-ID notation: `"sm"` (Spectrum Mill) or `"ph"` (Philosopher). Only used when `level = "ssc"`. |
| `id.type.out` | Output site-ID type: `"uniprot"`, `"refseq"`, or `"seqwin"`. Only used when `level = "ssc"`. |
| `acc.type` | Accession type of the `rid` in the input GCT: `"uniprot"`, `"refseq"`, or `"symbol"`. Only used when `level = "ssc"`. |
| `organism` | One of `"human"`, `"mouse"`, `"rat"`, `"zebrafish"`. Required only when mapping to UniProt (`level = "ssc"`, `id.type.out = "uniprot"`, `acc.type` in `c("refseq", "symbol")`). |
| `seqwin.col` | Column containing `\|`-separated flanking sequences; only used when `id.type.out = "seqwin"`. |
| `gene.col` | Column listing gene symbols; used for gene-centric levels and by `fix.gene.names`. |
| `fix.gene.names` | If `TRUE`, replace `gene.col` with gene names extracted from `fix.gene.names.column` via `fix.gene.names.regexpr` – a workaround for gene symbols corrupted by spreadsheet date-autoformatting (e.g. "SEPT10" becoming a date). |
| `fix.gene.names.column` | Column to extract corrected gene names from. |
| `fix.gene.names.regexpr` | Regular expression identifying the gene name within `fix.gene.names.column`. |
| `humanize.gene.names` | If `TRUE`, upper-case gene symbols (for mouse or rat data). Only used when `level = "gc"`. |
| `loc` | If `TRUE` (recommended), only fully localized PTM sites are kept. Only used when `level = "ssc"`. |
| `mode` | How multiple rows per gene are combined when `level = "gc"`, and how multiply-detected duplicate sites are combined when `level = "ssc"` (only `"mean"`, `"median"`, `"abs.max"` apply to `"ssc"`): `"mean"`, `"median"`, `"sd"` (keep the single row with the highest cross-sample standard deviation – i.e. the most variable/informative |

probe for that gene; "gc" only), "SGT" (subgroup-top: first Spectrum Mill subgroup per protein group; "gc" only), or "abs.max" (per sample, the most extreme value – for signed, log-transformed p-values).

| | |
|---|---|
| SGT.col | Column used to collapse to subgroup-top (mode = "SGT"). |
| mod.res | Modified residue(s), e.g. "S\|T\|Y" or "K". |
| mod.type | Modification type, e.g. "p" or "ac". |
| appenddim | See cmapR::write_gct(). |

## Value

The processed cmapR::GCT object (so it can be passed straight to ssGSEA3() without re-reading the file just written) – the actual path written to disk is attached as attr(result, "path").

---

# read_gene_sets *Import and merge one or more gene set databases*

---

## Description

Reads one or more GMT files, filters each by set size, and merges them into a single collection. If any gene set member carries a PTMsigDB ;u/;d directionality suffix, the whole collection is treated as PTMsigDB-style: directionality is split out into gs.direction and stripped from members.

## Usage

```
read_gene_sets(gene.set.databases, thres.min = 2, thres.max = 2000)
```

## Arguments

| | |
|---|---|
| gene.set.databases | Character vector of paths to GMT files. |
| thres.min | Minimum gene set size to keep. |
| thres.max | Maximum gene set size to keep. |

## Value

A list with gs, gs.names, gs.descs, size.G, and gs.direction (NULL for ordinary, non-directional gene sets).

`RefSeq2UniProt` *Map RefSeq accessions or gene symbols to UniProt accessions*

## Description

Uses the appropriate Bioconductor `org.*.eg.db` annotation package for the given `organism` to map identifiers to UniProt accessions.

## Usage

```
RefSeq2UniProt(
  ids,
  keytype = c("REFSEQ", "SYMBOL"),
  organism = c("human", "mouse", "rat", "zebrafish")
)
```

## Arguments

| | |
|---|---|
| `ids` | Character vector of accessions to map (RefSeq or gene symbols, per `keytype`). May contain version suffixes/site annotations after a `.` or `;`, which are stripped before querying. |
| `keytype` | One of `"REFSEQ"` or `"SYMBOL"`, matching the type of `ids`. |
| `organism` | One of `"human"`, `"mouse"`, `"rat"`, `"zebrafish"`. |

## Details

This replaces the previous auto-detection behavior (which sampled a few IDs and test-queried each of the human/mouse/rat/zebrafish annotation packages in turn to guess the organism). Auto-detection meant every call required all four `org.*.eg.db` packages to be installed, and could silently guess wrong on a small or ambiguous ID sample. Callers must now state the organism explicitly, and only the one relevant annotation package needs to be installed.

## Value

A list with elements `keytype`, `organism`, and `id.map` (a data.frame with columns `id`, `id.query`, `id.mapped`, `id.concat`). Entries with no match have `id.mapped = NA`.

`select_top_terms` *Select top-N FDR-significant gene sets by direction*

## Description

Picks gene sets that reach `fdr.max` in at least one sample, then keeps the `top_n` largest by absolute score (its most extreme per-sample value), separately for positive- and negative-scoring gene sets.

### Usage

```
select_top_terms(
  result,
  top_n = 10,
  direction = c("both", "pos", "neg"),
  fdr.max = 0.05
)
```

### Arguments

| | |
|---|---|
| result | An ssgsea_result, as returned by ssGSEA3(). |
| top_n | Number of gene sets to keep per direction. |
| direction | "both", "pos", or "neg". |
| fdr.max | Maximum (best-case, across samples) FDR-adjusted p-value for a gene set to be eligible. |

### Value

Character vector of selected gene set names.

---

## ssGSEA3 *Single-sample GSEA / PTM-SEA enrichment, with bootstrap significance*

---

### Description

Scores every sample in input.ds against every gene set in gene.set.databases, using a per-gene-set running-sum enrichment statistic (Subramanian et al. 2005; Abazeed et al. 2013) and, for output.score.type = "NES", a gene-level bootstrap to obtain a normalized enrichment score and p-value (see bootstrap_es_pvalue() for why this replaced the original permutation-based approach).

### Usage

```
ssGSEA3(
  input.ds,
  output.prefix,
  gene.set.databases,
  sample.norm.type = c("rank", "log", "log.rank", "none"),
  weight = 0,
  statistic = c("area.under.RES", "Kolmogorov-Smirnov"),
  output.score.type = c("NES", "ES"),
  nperm = 1000,
  min.overlap = 10,
  correl.type = c("rank", "z.score", "symm.rank"),
  global.fdr = FALSE,
  extended.output = TRUE,
  par = FALSE,
  spare.cores = 1,
  export.signat.gct = TRUE,
```

```
  param.file = TRUE,
  seed = NULL,
  verbose = TRUE,
  log.file = "run.log"
)
```

## Arguments

| | |
|---|---|
| input.ds | Path to a GCT file (first column: gene symbols or site ids), or an already-parsed cmapR::GCT object – e.g. the result of preprocessGCT(), to avoid re-reading it from disk. |
| output.prefix | File prefix for output GCT files. |
| gene.set.databases | Character vector of paths to GMT gene set database files. |
| sample.norm.type | See normalize_samples(). |
| weight | When 0, all genes are weighted equally; when > 0, actual (normalized) expression values are used and can shift the score – see correl.type. |
| statistic | "area.under.RES" or "Kolmogorov-Smirnov", see gsea_score(). |
| output.score.type | "NES" (bootstrap normalized enrichment score and p-value) or "ES" (raw enrichment score only, no bootstrapping, p.value is always 1). |
| nperm | Number of gene-level bootstrap replicates (ignored when output.score.type = "ES"). |
| min.overlap | Minimum number of a gene set's members that must be present in the data for it to be scored. |
| correl.type | "rank", "z.score", or "symm.rank" – only relevant when weight > 0, see gsea_score(). |
| global.fdr | If TRUE, FDR is computed once across all samples' pooled p-values; otherwise per sample. |
| extended.output | If TRUE, include per-signature overlap statistics as additional row annotations in the output GCTs. |
| par | If TRUE, score gene sets in parallel across parallel::detectCores() - spare.cores workers. |
| spare.cores | Number of CPU cores to leave idle when par = TRUE. |
| export.signat.gct | If TRUE, also write one GCT per signature (to a signature_gct/ subdirectory) containing that signature's members' original expression values. |
| param.file | If TRUE, write a <output.prefix>_parameters.txt record of the parameters used. |
| seed | Optional random seed, for reproducible bootstrap resampling. |
| verbose | If FALSE, suppress progress messages (the log file, if log.file is not NULL, is still written). |
| log.file | Path to a log file recording basic progress/timing; NULL to skip logging entirely. |

## Value

An object of class `ssgsea_result` (see `build_ssgsea_result()`); the same information is also written to `<output.prefix>-scores.gct`, `-pvalues.gct`, `-fdr-pvalues.gct`, and `-combined.gct` (plus, if `export.signat.gct`, one GCT per signature).

## Fixed limitation (directional/PTMsigDB gene sets)

earlier versions of this port could desync a directional (PTM-SEA/PTMsigDB `;u/;d`) gene set's position/direction bookkeeping inside `gsea_score()` whenever a bootstrap replicate happened to resample only *some* of a gene set's members – sometimes erroring out, sometimes silently contributing a wrong enrichment score to that replicate's bootstrap null. This is now fixed via `.align_geneset_to_direction()`, which keeps the two in lockstep regardless of which members a given replicate happens to include. The vulnerable construction traces back to code essentially unchanged since directionality support was added to the original ssGSEA2.0 in 2016, but it could never actually manifest there: ssGSEA2.0's gene-label *permutation* reorders every gene without ever dropping one, so its (fixed, computed-once) position/direction vectors never desync. It's specifically ssGSEA3.0's gene-level bootstrap resampling *with replacement* – which can and does drop individual genes from a given replicate – that exposes it; this is a real consequence of the bootstrap rework, not a latent defect ssGSEA2.0 use was ever exposed to.

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